

```
use vit
switched to db vit
db.sales.createIndex({
  item: 1
})
item_1
db.sales.getIndexes()
[
  { v: 2, key: { _id: 1 }, name: '_id_' },
  { v: 2, key: { item: 1 }, name: 'item_1' }
]
db.sales.find({
  item: "Americanos"
}).explain("executionStats")
{
  restoreState: 0,
  isEOF: 1,
  nss: 'vit.sales',
  docsExamined: 4,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 4,
    executionTimeMillisEstimate: 10,
    works: 5,
    advanced: 4,
    needTime: 0,
    needYield: 0,
```

```
saveState: 0,
restoreState: 0,
isEOF: 1,
nss: 'vit.sales',
keyPattern: {
  item: 1
},
indexName: 'item_1',
isMultiKey: false,
multiKeyPaths: {
  item: []
},
isUnique: false,
isSparse: false,
isPartial: false,
indexVersion: 2,
direction: 'forward',
indexBounds: {
  item: [
    '["Americanos", "Americanos"]'
  ]
},
keysExamined: 4,
seeks: 1,
dupsTested: 0,
dupsDropped: 0,
peakTrackedMemBytes: 0
}
```

```
}  
},  
queryShapeHash:  
'9383BE82251C8C530A346BE0E5F87D3AC8A4657C03B4033215799FE0667948A0',  
command: {  
  find: 'sales',  
  filter: {  
    item: 'Americanos'  
  },  
  '$db': 'vit'  
},  
serverInfo: {  
  host: 'LAPTOP-72DI8BHM',  
  port: 27017,  
  version: '8.3.2',  
  gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'  
},  
serverParameters: {  
  internalQueryFacetBufferSizeBytes: 104857600,  
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,  
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,  
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,  
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,  
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,  
  internalQueryProhibitBlockingMergeOnMongoS: 0,  
  internalQueryMaxAddToSetBytes: 104857600,  
  internalQueryFrameworkControl: 'trySbeRestricted',  
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
```

```
    },  
    ok: 1  
  }  
  executionStats: {  
    executionSuccess: true,  
    nReturned: 4,  
    totalKeysExamined: 4,  
    totalDocsExamined: 4  
  }
```

SyntaxError: Missing semicolon. (3:13)

```
1 | executionStats: {  
2 |   executionSuccess: true,  
> 3 |   nReturned: 4,  
   |       ^  
4 |   totalKeysExamined: 4,  
5 |   totalDocsExamined: 4  
6 | }
```

```
db.sales.insertMany([  
  {  
    "_id": 11,  
    "item": "Espresso",  
    "price": 12,  
    "size": "Short",  
    "quantity": 10  
  },  
  {  
    "_id": 12,
```

```
    "item": "Flat White",  
    "price": 14,  
    "size": "Tall",  
    "quantity": 8  
  }  
]);
```

```
{  
  acknowledged: true,  
  insertedIds: {  
    '0': 11,  
    '1': 12  
  }  
}
```

```
db.sales.createIndex({ date: 1 }, { unique: true });
```

MongoServerError[DuplicateKey]: Index build failed: 7b88844f-0ee6-4f1e-8851-ddb526e8c463: Collection vit.sales (4aea1a28-398c-4d7c-a839-90b68d20e1a6) :: caused by :: E11000 duplicate key error collection: vit.sales index: date_1 dup key: { date: null }

```
db.sales.insertMany([  
  {  
    _id: 11,  
    item: "Espresso",  
    price: 12,  
    size: "Short",  
    quantity: 10,  
    date: ISODate("2022-03-01T08:00:00Z")  
  },  
  {  
    _id: 12,
```

```
    item: "Flat White",
    price: 14,
    size: "Tall",
    quantity: 8,
    date: ISODate("2022-03-02T08:00:00Z")
  }
]);
```

MongoBulkWriteError: E11000 duplicate key error collection: vit.sales index: _id_ dup
key: { _id: 11 }

```
db.sales.aggregate([
  {
    $group: {
      _id: "$date",
      count: { $sum: 1 }
    }
  },
  {
    $match: {
      count: { $gt: 1 }
    }
  }
])
{
  _id: null,
  count: 2
}
db.sales.insertMany([
  {
```

```

    saleId: "S101",
    item: "Espresso",
    price: 12
  },
  {
    saleId: "S102",
    item: "Latte",
    price: 15
  }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a241f193eb3c19cb2da22c0'),
    '1': ObjectId('6a241f193eb3c19cb2da22c1')
  }
}
db.sales.createIndex(
  { saleId: 1 },
  { unique: true }
);

```

MongoServerError[DuplicateKey]: Index build failed: 9c5cf2b9-164a-4653-bc81-6c7f35e661a7: Collection vit.sales (4aea1a28-398c-4d7c-a839-90b68d20e1a6) :: caused by :: E11000 duplicate key error collection: vit.sales index: saleId_1 dup key: { saleId: null }

```
db.sales.countDocuments()
```

14

```
db.sales.drop()
```

true

```

db.sales.insertMany([
    { "_id" : 1, "item" : "Americanos", "price" : 5, "size": "Short", "quantity" : 22, "date" :
ISODate("2022-01-15T08:00:00Z") },
    { "_id" : 2, "item" : "Cappuccino", "price" : 6, "size": "Short","quantity" : 12, "date" :
ISODate("2022-01-16T09:00:00Z") },
    { "_id" : 3, "item" : "Lattes", "price" : 15, "size": "Grande","quantity" : 25, "date" :
ISODate("2022-01-16T09:05:00Z") },
    { "_id" : 4, "item" : "Mochas", "price" : 25,"size": "Tall", "quantity" : 11, "date" :
ISODate("2022-02-17T08:00:00Z") },
    { "_id" : 5, "item" : "Americanos", "price" : 10, "size": "Grande","quantity" : 12,
"date" : ISODate("2022-02-18T21:06:00Z") },
    { "_id" : 6, "item" : "Cappuccino", "price" : 7, "size": "Tall","quantity" : 20, "date" :
ISODate("2022-02-20T10:07:00Z") },
    { "_id" : 7, "item" : "Lattes", "price" : 25,"size": "Tall", "quantity" : 30, "date" :
ISODate("2022-02-21T10:08:00Z") },
    { "_id" : 8, "item" : "Americanos", "price" : 10, "size": "Grande","quantity" : 21,
"date" : ISODate("2022-02-22T14:09:00Z") },
    { "_id" : 9, "item" : "Cappuccino", "price" : 10, "size": "Grande","quantity" : 17,
"date" : ISODate("2022-02-23T14:09:00Z") },
    { "_id" : 10, "item" : "Americanos", "price" : 8, "size": "Tall","quantity" : 15, "date" :
ISODate("2022-02-25T14:09:00Z") }
]);
{
  acknowledged: true,
  insertedIds: {
    '0': 1,
    '1': 2,
    '2': 3,
    '3': 4,
    '4': 5,
    '5': 6,

```



```
    '6': 7,  
    '7': 8,  
    '8': 9,  
    '9': 10  
  }  
}
```

```
db.sales.createIndex(  
  { date: 1 },  
  { unique: true }  
);
```

date_1

```
db.sales.insertOne({  
  "_id": 11,  
  "item": "Espresso",  
  "price": 12,  
  "size": "Tall",  
  "quantity": 5,  
  "date": ISODate("2022-02-25T14:09:00Z")  
});
```

MongoServerError: E11000 duplicate key error collection: vit.sales index: date_1 dup
key: { date: new Date(1645798140000) }

```
db.sales.drop()
```

```
db.sales.insertMany([  
  { item: "Americanos", price: 5, size: "Short", quantity: 22 },  
  { item: "Americanos", price: 10, size: "Grande", quantity: 12 },  
  { item: "Cappuccino", price: 6, size: "Short", quantity: 12 }  
])
```

```
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6a2420b43eb3c19cb2da22c2'),
    '1': ObjectId('6a2420b43eb3c19cb2da22c3'),
    '2': ObjectId('6a2420b43eb3c19cb2da22c4')
  }
}
```

```
db.sales.createIndex({ item: 1 }, { unique: true })
```

MongoServerError[DuplicateKey]: Index build failed: 7372d630-57c8-48a8-b44b-bb75b5641c94: Collection vit.sales (42437e9a-0f8d-4de8-b0f3-695841c2612c) :: caused by :: E11000 duplicate key error collection: vit.sales index: item_1 dup key: { item: "Americanos" }

```
db.sales.deleteOne({
  item: "Americanos",
  price: 10
})
```

```
{
  acknowledged: true,
  deletedCount: 1
}
```

```
db.sales.createIndex({ item: 1 }, { unique: true })
```

item_1

```
db.sales.insertOne({
  item: "Americanos",
  size: "Grande",
  price: 10,
  quantity: 12
})
```

MongoServerError: E11000 duplicate key error collection: vit.sales index: item_1 dup
key: { item: "Americanos" }

```
db.sales.createIndex({
```

```
  item: 1,
```

```
  size: 1
```

```
})
```

```
item_1_size_1
```

```
...
```

```
winningPlan: {
```

```
  stage: 'FETCH',
```

```
  inputStage: {
```

```
    stage: 'IXSCAN',
```

```
    indexName: 'item_1_size_1',
```

```
    ...
```

```
  }
```

```
}
```

```
...
```

SyntaxError: Unexpected token (1:0)

```
> 1 | ...
```

```
  | ^
```

```
2 | winningPlan: {
```

```
3 |   stage: 'FETCH',
```

```
4 |   inputStage: {
```

```
db.sales.find({
```

```
  item: "Americanos",
```

```
  size: "Grande"
```

```
}).explain("executionStats")
```

```
{
  needTime: 0,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'vit.sales',
  keyPattern: {
    item: 1
  },
  indexName: 'item_1',
  isMultiKey: false,
  multiKeyPaths: {
    item: []
  },
  isUnique: true,
  isSparse: false,
  isPartial: false,
  indexVersion: 2,
  direction: 'forward',
  indexBounds: {
    item: [
      '["Americanos", "Americanos"]'
    ]
  },
  keysExamined: 1,
  seeks: 1,
  dupsTested: 0,
```

```
    dupsDropped: 0,
    peakTrackedMemBytes: 0
  }
}
},
queryShapeHash:
'303E34BF2FE739FF4FFA49F94F00D16DABCBBD4D0FBA695F323B1C759143D7B68',
command: {
  find: 'sales',
  filter: {
    item: 'Americanos',
    size: 'Grande'
  },
  '$db': 'vit'
},
serverInfo: {
  host: 'LAPTOP-72DI8BHM',
  port: 27017,
  version: '8.3.2',
  gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
```

```
internalQueryProhibitBlockingMergeOnMongoS: 0,
internalQueryMaxAddToSetBytes: 104857600,
internalQueryFrameworkControl: 'trySbeRestricted',
internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
db.sales.find({
  item: "Americanos"
}).explain("executionStats")
{
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
    isCached: false,
    stage: 'EXPRESS_IXSCAN',
    keyPattern: '{ item: 1 }',
    indexName: 'item_1'
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 18,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
```

```
executionStages: {
  isCached: false,
  stage: 'EXPRESS_IXSCAN',
  keyPattern: '{ item: 1 }',
  indexName: 'item_1',
  keysExamined: 1,
  docsExamined: 1,
  nReturned: 1
}
},
queryShapeHash:
'9383BE82251C8C530A346BE0E5F87D3AC8A4657C03B4033215799FE0667948A0',
command: {
  find: 'sales',
  filter: {
    item: 'Americanos'
  },
  '$db': 'vit'
},
serverInfo: {
  host: 'LAPTOP-72DI8BHM',
  port: 27017,
  version: '8.3.2',
  gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
```

```
internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
internalQueryFacetMaxOutputDocSizeBytes: 104857600,
internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
internalQueryProhibitBlockingMergeOnMongoS: 0,
internalQueryMaxAddToSetBytes: 104857600,
internalQueryFrameworkControl: 'trySbeRestricted',
internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
db.sales.find({
  size: "Grande"
}).explain("executionStats")
{
  size: {
    '$eq': 'Grande'
  }
},
nss: 'vit.sales',
direction: 'forward'
},
rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 0,
  executionTimeMillis: 0,
```



```
totalKeysExamined: 0,
totalDocsExamined: 2,
executionStages: {
  isCached: false,
  stage: 'COLLSCAN',
  filter: {
    size: {
      '$eq': 'Grande'
    }
  },
  nReturned: 0,
  executionTimeMillisEstimate: 0,
  works: 3,
  advanced: 0,
  needTime: 2,
  needYield: 0,
  saveState: 0,
  restoreState: 0,
  isEOF: 1,
  nss: 'vit.sales',
  direction: 'forward',
  docsExamined: 2
},
queryShapeHash:
'A0E3B1298BDACE1C98C70CDEE796D227734F0F585B3C0A67EE8783700A946410',
command: {
  find: 'sales',
```

```

filter: {
  size: 'Grande'
},
'$db': 'vit'
},
serverInfo: {
  host: 'LAPTOP-72DI8BHM',
  port: 27017,
  version: '8.3.2',
  gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
db.sales.createIndex({ item: 1, size: 1 })
item_1_size_1
db.sales.find({ item: "Americanos", size: "Grande" }).explain('executionStats');

```

```
{
  restoreState: 0,
  isEOF: 1,
  nss: 'vit.sales',
  docsExamined: 1,
  alreadyHasObj: 0,
  inputStage: {
    stage: 'IXSCAN',
    nReturned: 1,
    executionTimeMillisEstimate: 0,
    works: 2,
    advanced: 1,
    needTime: 0,
    needYield: 0,
    saveState: 0,
    restoreState: 0,
    isEOF: 1,
    nss: 'vit.sales',
    keyPattern: {
      item: 1
    },
    indexName: 'item_1',
    isMultiKey: false,
    multiKeyPaths: {
      item: []
    },
    isUnique: true,
    isSparse: false,
```

```
isPartial: false,
indexVersion: 2,
direction: 'forward',
indexBounds: {
  item: [
    ["Americanos", "Americanos"]
  ],
},
keysExamined: 1,
seeks: 1,
dupsTested: 0,
dupsDropped: 0,
peakTrackedMemBytes: 0
}
},
queryShapeHash:
'303E34BF2FE739FF4FFA49F94F00D16DABCBBD4D0FBA695F323B1C759143D7B68',
command: {
  find: 'sales',
  filter: {
    item: 'Americanos',
    size: 'Grande'
  },
  '$db': 'vit'
},
serverInfo: {
  host: 'LAPTOP-72DI8BHM',
```

```
port: 27017,
version: '8.3.2',
gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
db.sales.find({ item: "Americanos" }).explain('executionStats');
  queryHash: '54D41B5B',
  planCacheShapeHash: '54D41B5B',
  planCacheKey: 'C0BF8066',
  optimizationTimeMillis: 0,
  maxIndexedOrSolutionsReached: false,
  maxIndexedAndSolutionsReached: false,
  maxScansToExplodeReached: false,
  prunedSimilarIndexes: false,
  winningPlan: {
```

```
    isCached: false,
    stage: 'EXPRESS_IXSCAN',
    keyPattern: '{ item: 1 }',
    indexName: 'item_1'
  },
  rejectedPlans: []
},
executionStats: {
  executionSuccess: true,
  nReturned: 1,
  executionTimeMillis: 0,
  totalKeysExamined: 1,
  totalDocsExamined: 1,
  executionStages: {
    isCached: false,
    stage: 'EXPRESS_IXSCAN',
    keyPattern: '{ item: 1 }',
    indexName: 'item_1',
    keysExamined: 1,
    docsExamined: 1,
    nReturned: 1
  }
},
queryShapeHash:
'9383BE82251C8C530A346BE0E5F87D3AC8A4657C03B4033215799FE0667948A0',
command: {
  find: 'sales',
  filter: {
```

```

    item: 'Americanos'
  },
  '$db': 'vit'
},
serverInfo: {
  host: 'LAPTOP-72DI8BHM',
  port: 27017,
  version: '8.3.2',
  gitVersion: '89f54eb32e217d2f7bfcad50e5919f3033bb9611'
},
serverParameters: {
  internalQueryFacetBufferSizeBytes: 104857600,
  internalDocumentSourceGroupMaxMemoryBytes: 104857600,
  internalQueryMaxBlockingSortMemoryUsageBytes: 104857600,
  internalDocumentSourceSetWindowFieldsMaxMemoryBytes: 104857600,
  internalQueryFacetMaxOutputDocSizeBytes: 104857600,
  internalLookupStageIntermediateDocumentMaxSizeBytes: 104857600,
  internalQueryProhibitBlockingMergeOnMongoS: 0,
  internalQueryMaxAddToSetBytes: 104857600,
  internalQueryFrameworkControl: 'trySbeRestricted',
  internalQueryPlannerIgnoreIndexWithCollationForRegex: 1
},
ok: 1
}
db.sales.dropIndex({ item: 1 })
{ nIndexesWas: 3, ok: 1 }
db.sales.dropIndexes()
{

```

```
nIndexesWas: 2,  
msg: 'non-_id indexes dropped for collection',  
ok: 1  
}  
db.sales.getIndexes()
```